

Limitations

Due to computational resource constraints, we applied LoRA fine-tuning on the Qwen2.5-14B-Instruct model to validate our method, as discussed in Appendix H.2. Full-parameter fine-tuning may yield even better performance.

Future work will investigate larger LLM backbones to further validate the effectiveness of our method on models with greater parameter scales.

Moreover, although our method is already time-efficient, we plan to further improve efficiency by exploring lightweight training approaches, such as Low-Rank Adaptation (LoRA) and prefix tuning, during the fine-tuning of both the generation proxy model and the reward proxy model.

8 Ethics Statement

We adhere to the ACL Ethics Policy and all of our research is based on publicly available repositories and datasets. In the *RewardDS* framework, we uphold strict ethical standards to protect user privacy and ensure data security. The datasets used, covering medical QA, financial QA, and code generation domains, are publicly available and free of personally identifiable information, minimizing privacy risks. Our methodology does not access or reconstruct identifiable data, safeguarding individual privacy rights.

However, as our study involves multiple LLMs, such as Llama and Qwen, the findings may be influenced by the inherent biases, linguistic patterns, and assertiveness of these models.

References

- Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. 2016. [Deep learning with differential privacy](#). In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, CCS '16*, page 308–318, New York, NY, USA. Association for Computing Machinery.
- Sahar Abdelnabi, Kai Greshake, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. 2023. [Not what you’ve signed up for: Compromising real-world llm-integrated applications with indirect prompt injection](#). In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security, AISec 2023, Copenhagen, Denmark, 30 November 2023*, pages 79–90. ACM.
- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.
- Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, et al. 2021. Extracting training data from large language models. In *30th USENIX security symposium (USENIX Security 21)*, pages 2633–2650.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebggen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. 2021. [Evaluating large language models trained on code](#).
- Zeming Chen, Alejandro Hernández Cano, Angelika Romanou, Antoine Bonnet, Kyle Matoba, Francesco Salvi, Matteo Pagliardini, Simin Fan, Andreas Köpf, Amirkeivan Mohtashami, et al. 2023. Meditron-70b: Scaling medical pretraining for large language models. *arXiv preprint arXiv:2311.16079*.
- Christopher A Choquette-Choo, Florian Tramer, Nicholas Carlini, and Nicolas Papernot. 2021. Label-only membership inference attacks. In *International conference on machine learning*, pages 1964–1974. PMLR.

- Yihong Dong, Xue Jiang, Huanyu Liu, Zhi Jin, Bin Gu, Mengfei Yang, and Ge Li. 2024. [Generalization or memorization: Data contamination and trustworthy evaluation for large language models](#). In *Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and virtual meeting, August 11-16, 2024*, pages 12039–12050.
- Doubao. 2024. Doubao: Ai large model platform. Accessed: 2024-12-09.
- Cynthia Dwork and Aaron Roth. 2014. [The algorithmic foundations of differential privacy](#). *Found. Trends Theor. Comput. Sci.*, 9(3-4):211–407.
- James Flemings and Murali Annavaram. 2024. [Differentially private knowledge distillation via synthetic text generation](#). In *Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and virtual meeting, August 11-16, 2024*, pages 12957–12968. Association for Computational Linguistics.
- Jordan Frery, Roman Bredehocht, Jakub Klemsa, Arthur Meyre, and Andrei Stoian. 2025. Private lora fine-tuning of open-source llms with homomorphic encryption. *arXiv preprint arXiv:2505.07329*.
- Charlie Hou, Akshat Shrivastava, Hongyuan Zhan, Ryan Conway, Trang Le, Adithya Sagar, Giulia Fanti, and Daniel Lazar. 2024. Pre-text: Training language models on private federated data in the age of llms. *arXiv preprint arXiv:2406.02958*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [Lora: Low-rank adaptation of large language models](#). In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*.
- Yutong Hu, Quzhe Huang, Mingxu Tao, Chen Zhang, and Yansong Feng. 2024. [Can perplexity reflect large language model’s ability in long text understanding?](#) In *The Second Tiny Papers Track at ICLR 2024, Tiny Papers @ ICLR 2024, Vienna, Austria, May 11, 2024*.
- Siming Huang, Tianhao Cheng, Jason Klein Liu, Jiaran Hao, Liuyihan Song, Yang Xu, J. Yang, J. H. Liu, Chenchen Zhang, Linzheng Chai, Ruifeng Yuan, Zhaoxiang Zhang, Jie Fu, Qian Liu, Ge Zhang, Zili Wang, Yuan Qi, Yinghui Xu, and Wei Chu. 2024. [Opencoder: The open cookbook for top-tier code large language models](#).
- Alexey Kurakin, Natalia Ponomareva, Umar Syed, Liam MacDermed, and Andreas Terzis. 2023. [Harnessing large-language models to generate private synthetic text](#). *CoRR*, abs/2306.01684.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*.
- Haoran Li, Dadi Guo, Donghao Li, Wei Fan, Qi Hu, Xin Liu, Chunkit Chan, Duanyi Yao, Yuan Yao, and Yangqiu Song. 2024a. [Privlm-bench: A multi-level privacy evaluation benchmark for language models](#). *Preprint*, arXiv:2311.04044.
- Haoran Li, Xinyuan Zhao, Dadi Guo, Hanlin Gu, Ziqian Zeng, Yuxing Han, Yangqiu Song, Lixin Fan, and Qiang Yang. 2024b. [Federated domain-specific knowledge transfer on large language models using synthetic data](#). *CoRR*, abs/2405.14212.
- Yunxiang Li, Zihan Li, Kai Zhang, Ruilong Dan, Steve Jiang, and You Zhang. 2023. Chatdoctor: A medical chat model fine-tuned on a large language model meta-ai (llama) using medical domain knowledge. *Cureus*, 15(6).
- Chin-Yew Lin. 2004. [ROUGE: A package for automatic evaluation of summaries](#). In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain. Association for Computational Linguistics.
- Chris Yuhao Liu, Liang Zeng, Jiakai Liu, Rui Yan, Jujie He, Chaojie Wang, Shuicheng Yan, Yang Liu, and Yahui Zhou. 2024. Skywork-reward: Bag of tricks for reward modeling in llms. *arXiv preprint arXiv:2410.18451*.
- Qian Lou, Bo Feng, Geoffrey Charles Fox, and Lei Jiang. 2020. Glyph: Fast and accurately training deep neural networks on encrypted data. *Advances in neural information processing systems*, 33:9193–9202.
- Nils Lukas, Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, and Santiago Zanella Béguelin. 2023. [Analyzing leakage of personally identifiable information in language models](#). In *44th IEEE Symposium on Security and Privacy, SP 2023, San Francisco, CA, USA, May 21-25, 2023*, pages 346–363.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. [Self-refine: Iterative refinement with self-feedback](#). In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- Justus Matter, Zhijing Jin, Benjamin Weggenmann, Bernhard Schoelkopf, and Mrinmaya Sachan. 2022. [Differentially private language models for secure data sharing](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 4860–4873, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- MetaAI. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

- Jianmo Ni, Gustavo Hernández Ábrego, Noah Constant, Ji Ma, Keith B. Hall, Daniel Cer, and Yinfei Yang. 2022. [Sentence-t5: Scalable sentence encoders from pre-trained text-to-text models](#). In *Findings of the Association for Computational Linguistics: ACL 2022, Dublin, Ireland, May 22-27, 2022*, pages 1864–1874.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. [Direct preference optimization: Your language model is secretly a reward model](#). In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- Aleksandr Romanov, Anna Kurtukova, Anastasia Fedotova, and Roman Meshcheryakov. 2019. Natural text anonymization using universal transformer with a self-attention. In *Proceedings of the III International Conference on Language Engineering and Applied Linguistics (PRLEAL-2019), Saint Petersburg, Russia*, pages 22–37.
- Robin Staab, Mark Vero, Mislav Balunovic, and Martin T. Vechev. 2024. [Large language models are advanced anonymizers](#). *CoRR*, abs/2402.13846.
- Latanya Sweeney. 1997. [Guaranteeing anonymity when sharing medical data, the datafly system](#). *Proceedings: a conference of the American Medical Informatics Association. AMIA Fall Symposium*, pages 51–5.
- Wenhao Wang, Xiaoyu Liang, Rui Ye, Jingyi Chai, Siheng Chen, and Yanfeng Wang. 2024. [Knowledge: Privacy-preserving synthetic text generation with knowledge distillation from server](#). In *Conference on Empirical Methods in Natural Language Processing*.
- Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. 2022. [Self-instruct: Aligning language models with self-generated instructions](#). In *Annual Meeting of the Association for Computational Linguistics*.
- Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhajan Kambar, David Rosenberg, and Gideon Mann. 2023. [Bloomberggpt: A large language model for finance](#). *arXiv preprint arXiv:2303.17564*.
- Chulin Xie, Zinan Lin, Arturs Backurs, Sivakanth Gopi, Da Yu, Huseyin A. Inan, Harsha Nori, Haotian Jiang, Huishuai Zhang, Yin Tat Lee, Bo Li, and Sergey Yekhanin. 2024. [Differentially private synthetic data via foundation model apis 2: Text](#). In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net.
- An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. 2024a. [Qwen2 technical report](#). *CoRR*, abs/2407.10671.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. 2024b. [Qwen2. 5](#) technical report. *arXiv preprint arXiv:2412.15115*.
- Samuel Yeom, Irene Giacomelli, Matt Fredrikson, and Somesh Jha. 2018. Privacy risk in machine learning: Analyzing the connection to overfitting. In *2018 IEEE 31st computer security foundations symposium (CSF)*, pages 268–282. IEEE.
- Zhichao You, Xuewen Dong, Ke Cheng, Xutong Mu, Jiaxuan Fu, Shiyang Ma, Qiang Qu, and Yulong Shen. 2025. [Prifft: Privacy-preserving federated fine-tuning of large language models via function secret sharing](#). *arXiv preprint arXiv:2503.03146*.
- Da Yu, Arturs Backurs, Sivakanth Gopi, Huseyin Inan, Janardhan Kulkarni, Zinan Lin, Chulin Xie, Huishuai Zhang, and Wanrong Zhang. 2023. Training private and efficient language models with synthetic data from llms. In *Socially Responsible Language Modelling Research*.
- Da Yu, Peter Kairouz, Sewoong Oh, and Zheng Xu. 2024. [Privacy-preserving instructions for aligning large language models](#). In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*.
- Da Yu, Saurabh Naik, Arturs Backurs, Sivakanth Gopi, Huseyin A. Inan, Gautam Kamath, Janardhan Kulkarni, Yin Tat Lee, Andre Manoel, Lukas Wutschitz, Sergey Yekhanin, and Huishuai Zhang. 2022. [Differentially private fine-tuning of language models](#). In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net.
- Xiang Yue, Huseyin Inan, Xuechen Li, Girish Kumar, Julia McAnallen, Hoda Shajari, Huan Sun, David Levitan, and Robert Sim. 2023. [Synthetic text generation with differential privacy: A simple and practical recipe](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1321–1342, Toronto, Canada. Association for Computational Linguistics.
- Ziqian Zeng, Jianwei Wang, Junyao Yang, Zhengdong Lu, Huiping Zhuang, and Cen Chen. 2024. [Privacyre-store: Privacy-preserving inference in large language models via privacy removal and restoration](#). *arXiv preprint arXiv:2406.01394*.
- Boyu Zhang, Hongyang Yang, and Xiao-Yang Liu. 2023. [Instruct-fingpt: Financial sentiment analysis by instruction tuning of general-purpose large language models](#). *FinLLM Symposium at IJCAI 2023*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging llm-as-a-judge with mt-bench and chatbot arena](#). In *Advances in Neural Information Processing Systems*